26900104 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECHE300 - QUANTUM MECHANICS, SPECTROSCOPY & GROUP THEORY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Write the time-independent Schrodinger wave equation, in cartesian coordinates [U] / [CO1]
2. Sketch the shapes of $1s$ and $2s$ orbitals. [K] / [CO1]
3. Sketch the wavefunction corresponding to quantum number $n = 2$ for a particle in a one - dimensional box with infinite potential walls. [A] / [CO1]
4. Write the expression for the number of vibrational degrees of freedom of a nonlinear molecule. [U] / [CO2]
5. Explain the different regions of electromagnetic spectrum used in spectroscopy. [U] / [CO2]
6. What is meant by a harmonic oscillator in vibrational spectroscopy? [U] / [CO2]
7. Show how many spin states are possible for Carbon-13. [U] / [CO3]
8. What is meant by a singlet electronic state? How does it differ from a triplet state? [K] / [CO3]

9. List two types of unsaturated functional groups that exhibit magnetic anisotropy. [K] / [CO3]

10. Identify the point groups of the molecules Cyclohexane (boat) and *Cis*-$[PtCl_2Br_2]$. [A] / [CO4]

11. Identify the symmetry operations generated by the C_3 axis. [A] / [CO4]

12. Identify the point groups of a centrosymmetric linear molecule and a non-centrosymmetric linear molecule, giving one example of each. [A] / [CO4]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

13. What are spherical polar coordinates? Write the time - independent Schrödinger equation for hydrogen atom in terms of spherical polar coordinates. [U] / [CO1]

14. What are quantum mechanical operators. Discuss their significance. [U] / [CO1]

15. Explain how rotational spectroscopy can be applied to determine the bond length of a diatomic molecule. [U] / [CO2]

16. What are the limitations of Beer-Lambert's law ? [K] / [CO2]

17. Explain the different types of electronic transitions involving σ, π and n molecular orbitals. Give suitable examples for each transition. [U] / [CO3]

18. Explain the Laporte selection rule and discuss its significance in electronic spectra of molecules. [U] / [CO3]

19. Explain the properties of a mathematical group. [U] / [CO4]

20. Using the symmetry operations and group multiplication table of C_{2V}, determine whether the point group is Abelian. Justify your answer. [A] / [CO4]

[5x6 = 30]

Part C

Short Essay

Answer any **2** questions

Each question carries **10** marks

21. Elaborate on the black body spectrum and the Planck distribution law. Discuss their significance in development of quantum mechanics. [U] / [CO1]

22. Explain the concept of anharmonicity in molecular vibrations and show how the dissociation energy of a diatomic molecule can be determined using the Morse potential. [U] / [CO2]

23. Explain the physical and electronic mechanisms of shielding and deshielding. Detail how these localized phenomena modify the Larmor frequency. [U] / [CO3]

24. Explain matrix representation of symmetry operations. Write down the matrix representation for E, i, S_n and σ. [U] / [CO4]

[10x2 = 20]